

Exhibit 1: Statement of Objectives (SOO)

Date: 4/8/2026

1.0 Purpose

This Statement of Objectives (SOO) establishes the top-level objectives for the Mars Telecommunications Network (MTN) procurement, which will be composed of three phases: (1) development through implementation, (2) launch and post-launch check-out, and (3) transit to Mars through relay commissioning. These objectives, detailed in Section 4.0, MTN Core Contract Objectives (CCO), define the Government's desired outcomes for this effort. The SOO format provides Offerors with maximum flexibility to develop cost-effective solutions and propose innovative approaches to achieve these objectives across all program phases.

Offerors shall develop and propose a Contract Performance Work Statement (PWS) that demonstrates how their proposed solutions will satisfy the MTN objectives while meeting all requirements specified in the Request for Proposal (RFP). The Offeror's proposed approach must comply with the requirements documented in Attachment B, Mars Telecommunications Network (MTN) Objectives and Requirements; Attachment F, Data Requirement Deliverables; and Enclosure 1, Draft Performance Work Statement (PWS).

Upon contract award, the selected Offeror's proposed PWS will be incorporated into the contract as Attachment A, MTN Performance Work Statement (PWS).

2.0 Scope

The objectives for the MTN procurement are described in Section 4.0, MTN Core Contract Objectives (CCO). The Offeror's approach to providing insight is a vital aspect of successful contract performance throughout all objectives stated in the SOO. The insight NASA achieves is essential in the mitigation of risk during execution of the contract.

3.0 Background

The One Big Beautiful Bill Act (OBBBA), officially Public Law (PL) 119-21, was signed into law on July 4, 2025. The law mandates that NASA use a competitive process and award a firm fixed-price contract with a U.S. commercial provider for a high-performing Mars telecommunications capability. NASA has designated this effort as the Mars Telecommunications Network (MTN). PL 119-21 requires delivery no later than December 31, 2028, with the launch window for ballistic transfer trajectories to Mars beginning in

November 2028 to meet the calendar year 2030 requirement for readiness to support relay operations.

The high-level objectives of the MTN project are twofold: (1) robustify the current Mars Relay Network (MRN) while paving the way to a new relay network architecture at Mars and (2) meet the needs of future Mars missions with focus on both exploration and science objectives. Given the state of the current MRN, there is an urgent need to satisfy objective (1). As such MTN will focus on solutions that meet this urgency with the objective to put the next relay asset in its operational orbit at Mars and begin relay service no later than CY30.

The MTN requirement encompasses full-scale development, integration, launch, and on-orbit operations to provide robust, continuous communications for future Mars surface and orbital missions and eventual human exploration missions. The network must support legacy users of the existing Mars Relay Network (MRN), including operational rovers, while also accommodating new users anticipated to operate at Mars through 2035. MTN capabilities will include communications and data exchange, positioning and navigation services (Doppler, range, and time transfer), autonomous operations, onboard processing, and extended mission duration.

MTN is designated a medium-priority Category 2 project, requiring heightened management agility, elevated risk tolerance to meet urgent funding and schedule constraints. Performance requirements are defined using threshold and objective values to support rapid source selection and to enable evaluation of value associated with performance exceeding thresholds. The objectives for this procurement, detailed in Section 4.0, are designed to establish an end-to-end Mars telecommunications and navigation relay capability that serves current operational missions, supports Entry, Descent, and Landing (EDL) demonstration missions, and provides the infrastructure necessary for the next generation of Mars exploration.

The requirements in Attachment B establish both threshold and objective performance levels for the MTN. Threshold requirements represent the minimum acceptable capability the Government will accept, while objective requirements represent the desired performance goals the Government values and aspires to achieve. Offerors are encouraged to propose system capabilities that exceed threshold requirements where technically feasible and programmatically sound. The Government places value on above-threshold performance, and Offerors should refer to the proposal instructions for guidance on how such capabilities should be reflected in their proposed PWS in order to be considered during evaluation.

4.0 MTN Core Contract Objectives (CCO)

CCO01 Detailed Milestones Definition and Payment Schedule:

Relevant NASA project management requirements documents (NPR 7120.5 and NID 7120.148) are provided for reference on the standard NASA project life cycle and expectations. Due to the compressed schedule, the Offeror may utilize its own life cycle plans and practices.

The Offeror will define, manage, and implement plans and processes associated with all phases of the MTN mission. The execution of these processes will assure that MTN is developed, produced, and operated in an efficient and transparent manner throughout the life cycle of the system. The Offeror will develop and propose a comprehensive schedule that defines all major project milestones, technical reviews, deliverables, and decision points necessary to achieve MTN certification of flight readiness and operational readiness. The schedule must support hardware delivery no later than December 31, 2028 and full operational readiness at Mars by the end of calendar year 2030. The milestone schedule will clearly identify critical path activities, dependencies, and integration points with NASA processes and reviews.

The Offeror will propose a payment schedule tied to the accomplishment of specific, verifiable milestones and deliverables. The payment structure will demonstrate clear linkage between payments and the completion of objective criteria that reduce program risk and demonstrate progress toward hardware delivery, certification and operational capability.

NASA is moving forward with urgency, and the NASA MTN Project strives to meet an aggressive schedule. With a hardware delivery deadline of calendar year 2028, NASA is acutely aware of the challenge that schedule presents. The Offeror will describe their approach to meeting this deadline, managing schedule, and managing associated risks, such as supply chain challenges and unique long-lead items necessary for this project. The approach will identify schedule margin allocation, risk mitigation strategies, and recovery options to address potential delays while maintaining the required delivery date.

CCO02 Requirements Flow Down and Validation:

The Offeror will establish and implement a comprehensive requirements management process that flows down the high-level requirements specified in Attachment B, *Mars*

Telecommunications Network (MTN) Objectives and Requirements, into lower-level design, performance, and verification requirements. This flow down will address all requirements categories defined in Attachment B. The Offeror's requirements management approach will ensure complete traceability from top-level objectives through all levels of system, subsystem, and component requirements.

The Offeror will define and implement a requirements validation process to ensure that all derived requirements are necessary, achievable, verifiable, and traceable to the parent requirements in Attachment B. The validation process will identify and resolve conflicts, gaps, or ambiguities in the requirements set and will demonstrate that the complete set of requirements, when satisfied, will meet the MTN objectives stated in this SOO.

The Offeror will maintain a requirements database or equivalent system that captures all requirements, their relationships, validation status, and verification approach throughout the program lifecycle. The Offeror will provide NASA with visibility into the requirements management process and status to support insight and oversight activities defined in CCO05.

CCO03 Verification Plans:

The Offeror will develop and implement comprehensive verification plans that demonstrate how all requirements specified in Attachment B, *Mars Telecommunications Network (MTN) Objectives and Requirements*, and all derived lower-level requirements will be verified throughout the MTN development lifecycle. The verification approach will define the methods (analysis, inspection, demonstration, and test) for each requirement and will establish the verification sequence, environments, success criteria, and responsible organizations for all verification activities.

The Offeror will address the core philosophy of the verification approach and trades that are in play to balance the tension between a rigorous verification program and the aggressive delivery schedule. The Offeror will identify a process by which deviations and waivers to the plan will be approved by NASA.

The Offeror's verification plans will address all phases of the program including design verification, qualification testing, acceptance testing, integration and test, launch readiness verification, and on-orbit checkout and commissioning. The verification approach will demonstrate how the Offeror will confirm that the MTN design and

construction activities achieve the required performance, reliability, safety, and mission assurance objectives.

The verification plan will provide NASA with visibility into verification methods, criteria, and results, and will include appropriate insight into verification activities to support the certification process and ensure that the MTN meets all performance objectives prior to operational deployment. The Offeror will provide access to key verification events and test activities as appropriate to enable Government oversight and risk assessment.

CCO04 Standards:

The Offeror will establish and implement relevant technical standards (e.g., design and construction, environmental verification) that ensure technical excellence and mission success for the MTN. NASA technical standards and procedural requirements are one means of assuring mission success and are a pillar of technical excellence at NASA. They often represent years of NASA lessons learned and are updated to reflect best practices. NASA also acknowledges that industry has developed and adopted standards that are regularly followed for their applications.

The Offeror will describe their approach to utilization of standards and cite those they will make applicable and implement on this project such that the Government can assess deviations to NASA standards. The approach will address how industry standards and best practices will be applied and will identify where NASA standards are referenced in Attachment B, Section titled, *“Reference and Applicable Documents”* will be adopted, tailored, or alternatively satisfied through equivalent industry standards.

The Offeror will demonstrate how their selected standards and technical processes will ensure the MTN achieves the required performance, reliability, safety, and mission assurance objectives. The approach will provide NASA with visibility into the standards baseline and any proposed deviations or tailoring, enabling appropriate Government insight and risk assessment throughout the design, development, and construction phases.

CCO05 Risk Assessment and Recommended Mitigations:

The Offeror will establish and implement a comprehensive risk management process that identifies, assesses, tracks, and mitigates technical and programmatic risks throughout the MTN lifecycle. The Offeror will identify and implement a process to determine and mitigate risk throughout the life cycle. The Offeror will involve the NASA MTN project in the

discussion of such risks and seek NASA concurrence on any deviations from plan. The risk management approach will address all phases of the program including design, development, integration, test, launch, and operations. The Offeror will develop risk mitigation strategies for identified risks and will provide recommended mitigations with associated implementation plans, costs, and schedule impacts.

NASA understands that some risks are worth taking. However, no risk should ever be accepted blindly or without an appropriate level of information and data to support a decision. Balancing risk is a pillar of technical excellence at NASA. The MTN Project intends to gather information and data throughout the project lifecycle via technical insight as the vendor is performing design, development, testing, and integration activities. The Offeror will describe their approach to ensuring NASA has access to information and data necessary to make informed decisions related to risk management. The Offeror will describe how NASA will have insight into their risk management process and how risks will be elevated for NASA support in mitigation when appropriate.

The Offeror's risk management approach will provide visibility into risk status, trends, and mitigation effectiveness. The approach will include processes for timely communication of emerging risks, changes in risk posture, and situations where NASA involvement or decision-making is required. Given the aggressive schedule constraints and launch window requirements for MTN, the Offeror will specifically address risk management strategies related to schedule compression, supply chain vulnerabilities, long-lead items, and critical path activities.

CCO06 NASA Engagement Plan - Insight and Oversight:

The Offeror will develop and propose a comprehensive NASA engagement plan that defines how the Government will be integrated into the MTN project to provide an appropriate blend of insight and oversight throughout all phases of the project. The Offeror's engagement plan will demonstrate how NASA will maintain visibility into project progress, technical decisions, risk management, and quality assurance while respecting the Offeror's role as the prime developer and operator of the MTN.

The Offeror will define their proposed approach to NASA engagement across all project activities including, but not limited to, design reviews, technical interchange meetings, testing and verification activities, risk management processes, certification milestones, problem and anomaly investigations, and operational readiness assessments. The Government is not prescribing specific engagement requirements or processes; rather, the Offeror has flexibility to propose innovative approaches that balance NASA's need for

informed decision-making with efficient project execution. Examples of engagement mechanisms may include technical interchange meetings, access to facilities and testing activities, data access and reporting structures, participation in key reviews and decision gates, and collaboration forums, but the Offeror is encouraged to propose the engagement approach that best suits their project implementation strategy.

The Offeror will describe how their NASA engagement plan integrates with and supports the requirements validation process (CCO02), verification plans (CCO03), verification of design and construction standards (CCO04), and risk-informed decision making (CCO05). The engagement approach will ensure that NASA has timely access to the information and data necessary to make informed decisions, assess project health, and provide support when needed. The plan will identify key touchpoints, milestones, gates, and decision points where Government engagement is proposed, and will describe the level and type of NASA involvement at each stage. The Offeror's approach will demonstrate a coherent strategy that enables NASA to fulfill its oversight responsibilities while allowing the Offeror the flexibility to execute the project efficiently and meet the aggressive schedule requirements.

CCO07 IT Systems Security Approach:

The Offeror will develop and implement a comprehensive IT systems security approach that protects all MTN information technology systems, networks, data, and interfaces throughout the project lifecycle. The IT security approach will address ground systems, mission operations centers, command and control systems, data handling and storage systems, and communications interfaces with NASA networks and facilities. The Offeror's approach will ensure the confidentiality, integrity, and availability of MTN systems and data against cybersecurity threats and vulnerabilities.

The Offeror will describe their IT systems security framework and how it aligns with industry best practices and applicable standards such as National Institute of Standards and Technology (NIST) cybersecurity guidelines and NASA IT security standards. The approach will address security controls, risk management, vulnerability assessment, incident response, and continuous monitoring throughout system development, integration, test, launch, and operational phases.

The Offeror will define how their IT systems security approach interfaces with NASA IT security requirements and processes. The approach will provide NASA with visibility into the security posture of MTN IT systems and will establish processes for coordinating

security-related activities, reporting security incidents, and implementing security updates or patches during operational phases.

CCO08 MTN Technical Approach (Communications and Navigation):

The Offeror will develop and propose a comprehensive technical approach that addresses the MTN mission objectives and demonstrates how the proposed system architecture will achieve the required capabilities. The technical approach will address the mission objectives specified in Attachment B. The Offeror's technical approach will demonstrate how the proposed solution meets or exceeds the performance requirements in all four objective areas while supporting the aggressive delivery and launch schedule.

The Offeror will describe their approach to the communication and Positioning, Navigation, and Timing (PNT) architecture, including both ground and space architecture elements. The description will address how the ground systems and space segment work together to provide the required communications relay, data exchange, and navigation services. The architecture approach will identify all major system elements, their functions, interfaces, and how they integrate to form an end-to-end capability that serves both legacy Mars missions and future users through 2035.

CCO09 MTN Technical Approach (Transit to Mars and Delta-V):

The Offeror will describe their approach to transit to Mars and delta-V margin management. The approach will address the propulsion strategy, trajectory design, and explain the rationale for the selected transit method. The Offeror will describe their approach to delta-V margin allocation to allow for orbital maneuvers at Mars over the operational lifetime, addressing how sufficient margin will be maintained to support mission flexibility, orbit maintenance, and potential repositioning requirements throughout the operational phase.

CCO10 MTN Technical Approach (Extensibility):

The Offeror will describe their approach to extensibility and long-term system viability beyond 2035. Given the significant investment in MTN infrastructure, the Offeror will address how the system design supports extended operational life and future expansion or enhancement. The approach will describe interoperability strategies aligned with emerging standards such as the LunaNet Interoperability Specification (LNIS), managed by NASA, ESA, and JAXA, which provides a framework for creating a cooperative "network of networks" to ensure long-term viability. The Offeror will explain how their architecture

supports future interoperability with other deep space networks and enables MTN to serve as part of a broader space communications infrastructure beyond its initial operational period.